Discrete Fourier Transform

- Practical Tool for Fourier Analysis

Summary of Fourier Transform

	Signal		Spectrum
1.	CT Signal $x_c(t)$ Non periodic	CTFT	$X_c(\Omega)$ @ Continuous frequency Ω. $-\infty < \Omega < \infty$
2.	CT Signal $x_p(t)$ Periodic fundamental period T_0 sec.	CTFS	Fundamental freq $= \frac{2\pi}{T_0} = \Omega_0$ Harmonics $= k\Omega_0$ $k \in Z$ Spectrum is Discrete.
3.	DT Signal $\{x(n)\}$ not periodic	DTFT	$X(\omega)$; $-\pi < \omega < \pi$ @ Freq. ω is continuous Spectrum is periodic $X(\omega) = X(\omega + 2\pi)$
4.	DT Signal $\{x(n)\}$ Periodic $x(n) = x(n+N)$	DFT DTFS	$X(\omega)$ => Periodic AND Discrete

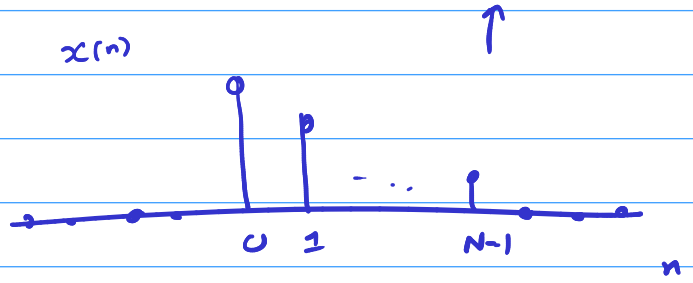
(DFT) (DTFS)

Discrete Fourier Transform & Discrete time Fourier Series

Suppose we have finite length signal

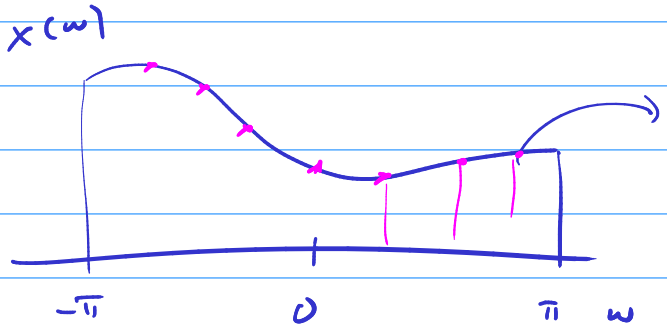
$\{x(n)\}$ has length N

$\{x(n)\} = \{x(0), x(1), \ldots, x(N)\}$



DTFT of $\{x(n)\}$

$$X(w) = \sum_{n=0}^{N-1} x(n) e^{-jwn}$$



$-\pi < w \leq \pi$

N → samples of spectrum

(DFT)

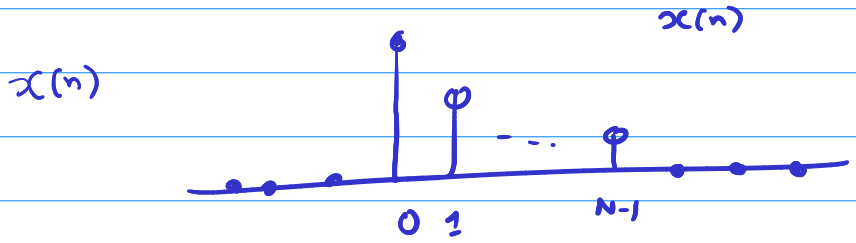
Discrete Fourier Transform of Discrete Time Fourier Series (DTFS)

Consider a finite length signal

$$x(n) = \{ x(0), \cdots, x(N-1) \}$$

$\uparrow$ $n=0$

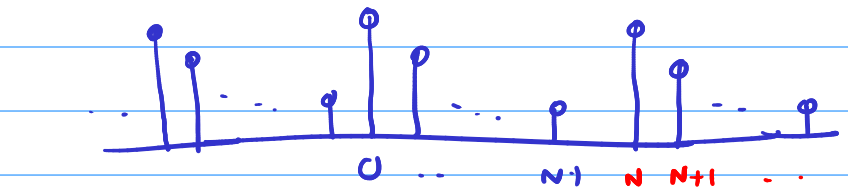
length N



Consider periodic repetition of $x(n)$

$$\tilde{x}(n) = x(n) \quad ; \quad n = 0, \cdots, N-1$$

$$\tilde{x}(n+N) = \tilde{x}(n) \quad \text{for all } n$$



Periodic signal have discrete spectrum

Modulo N arithmetic

$$\tilde{x}(n) = x(n \bmod N)$$

$n \bmod N \triangleq$ remainder when n is divided by N

quotient

$-1 = (-1)N + (N-1)$ (remainder)

$$n \bmod N \in \{0, \ldots, N-1\}$$

$0 \bmod N = 0$

$1 \bmod N = 1$

$-1 \bmod N = N-1$ & so on.

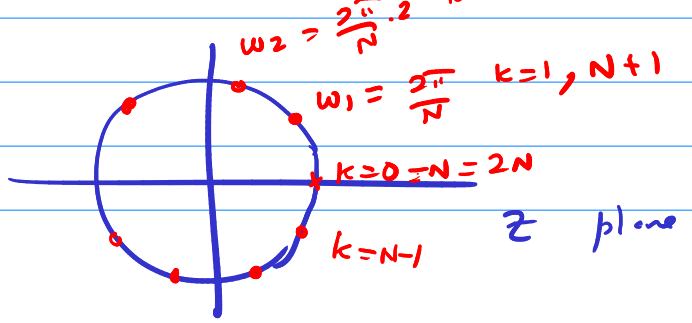
Periodic signal $\tilde{x}(n)$

Fundamental Period $= N$

angular Fundamental freq $= \frac{2\pi}{N}$ radians/sample

k^{th} harmonic $= k\frac{2\pi}{N} = \omega_k$

Corresponding Complex exponential $e^{j\omega_k n} = e^{j\frac{2\pi}{N}kn}$



There are only finite number of harmonics

$$\omega_k = k\frac{2\pi}{N} \qquad e^{j\omega_k n}$$

$$k \in \{0, 1, \cdots, N-1\}$$

$$\omega_k \equiv \omega_{k \bmod N}$$

Discrete time Fourier Series

$$\tilde{x}(n) = \sum_{k=0}^{N-1} \frac{\tilde{a}_k}{N} e^{j\omega_k n} \quad ; n \in \mathbb{Z}$$

$$\tilde{a}_k = \sum_{n=0}^{N-1} \tilde{x}(n)\, e^{-j\omega_k n}$$

Proof.

$$\tilde{x}(n) = \sum_{k=0}^{N-1} \frac{\tilde{a}_k}{N} e^{+j\omega_k n}$$

$$\omega_l = \frac{l 2\pi}{N}$$

$$\tilde{x}(n)\, e^{-j\omega_l n} = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{a}_k\, e^{j(\omega_k - \omega_l) n}$$

$$\sum_{n=0}^{N-1} \tilde{x}(n)\, e^{-j\omega_l n} = \frac{1}{N} \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} \tilde{a}_n\, e^{j(k-l)\frac{2\pi}{N} n}$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} \tilde{a}_k \underbrace{\sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-l)n}}$$

$$=$$

$= \tilde{a}_l$

$$\sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\omega_l n} = \tilde{a}_l$$

$$= \begin{cases} N & \text{if } k=l \\ 0 & \text{if } k \neq l \end{cases}$$

l^{th} harmonic Fourier Series coeff.

$x(n)$ coincides with $\tilde{x}(n)$ for $n = 0, \cdots, N-1$

We can rewrite DTFS as

$$\text{DFT} \quad X[k] = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn} \qquad k = 0, \cdots, N-1$$

$$\text{IDFT} \quad x(n) = \sum_{k=0}^{N-1} \frac{X[k]}{N} e^{+j\frac{2\pi}{N}kn} \quad ; \quad n = 0, 1, \cdots, N-1$$

Remarks

$X[k] \Rightarrow$ DFT coefficients

There are N DFT coeff.

N length DT signal

$\Updownarrow$ N Point DFT

N DFT coefficients

- Inverse DFT expression is inherently periodic

If we allow n to be any integer instead of restricting to $\{0, \ldots, N-1\}$

the IDFT expression will give a periodic signal.

Consider DT Signal $\{x(n)\}$

We can find DTFT

$$\{x(n)\} \longleftrightarrow X(\omega) \quad \text{spectrum} \quad -\pi < \omega \leq \pi$$

$$\{x(n)\} \xleftrightarrow{N\text{ pt DFT}} X[k] \quad ; \quad k = 0, \ldots, N-1 \quad \text{DFT coeff.}$$

DFT bin freq.

k^{th} = k^{th} harmonic fun.

$$\omega_k = k \frac{2\pi}{N}$$

Relation between spectrum $X(\omega)$ & DFT coeff $X[k]$

Case A

$\{x(n)\}$ is finite length signal with length N

$n=0$

$$\{x(n)\} = \{x(0), \ldots, x(N-1)\}$$

length N

$$\text{DTFT} \quad X(\omega) = \sum_{n=0}^{N-1} x(n)\, e^{-j\omega n}$$

$$0 < \omega < 2\pi$$

$$\text{DFT} \quad X[k] = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}kn}$$

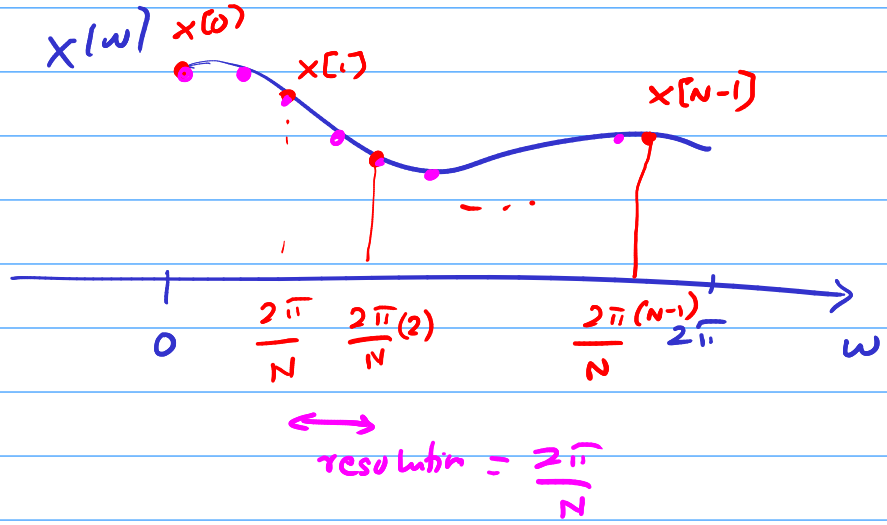
$$k = 0, 1, \ldots, N-1$$

$$X[k] = X(\omega)\Big|_{@\ \omega = \frac{2\pi}{N}k}$$

Time for finite length signal

Sampling Spectrum at bin frequency (k^{th} harmonics)

$L = 2N$



Case B $\{x(n)\}$ is finite length signal of length N

$$\{x(n)\} = \{x(0), \ldots, x(N-1)\}$$

Zero padding

Add $L-N$ zeros

$$\{x(n)\} = \{x(0), \ldots, x(N-1), \underbrace{0, 0, \ldots 0}_{L-N \text{ zeros}}\}$$

Length L array, $L > N$

L point DFT

$$X[m] = \sum_{n=0}^{L-1} x(n)\, e^{-j\frac{2\pi}{L}mn}$$

m^{th} bin freq. for L pt DFT

$$= X(\omega) \Big|_{@\ \omega = \frac{2\pi}{L}m}$$

resolution $= \frac{2\pi}{L}$ (improved)

Case C $\{x(n)\}$ is not finite length

(length > N)

$\{x(n)\} \Rightarrow$ length > N

Take first N samples

$\{x(0), \ldots, x(N-1)\}$

$\Downarrow$

N pt DFT

$\{X[k];\ k = 0, \ldots, N-1\}$

$$\{x(n)\} \longleftrightarrow X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$

Truncated signal

$$x_T[n] = \begin{cases} x(n) ; & n = 0, \ldots, N-1 \\ 0 ; & \text{else} \end{cases}$$

$\Downarrow$

Spectrum

$$X_T(\omega) = \sum_{n=0}^{N-1} x(n)\, e^{-j\omega n}$$

$X[k]$ = DFT of $\{x_T[n]\}$

$X[k] \Rightarrow$ Samples of spectrum of finite length signal $x_T[n]$

$$X[k] = X_T(\omega)\Big|_{@\ \omega = \frac{2\pi}{N}k}$$

Need to find relation between

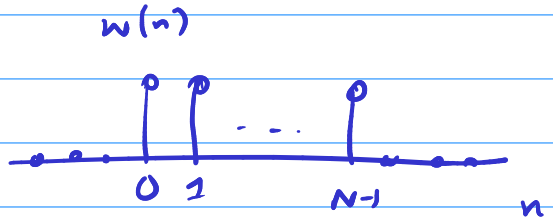
$X(\omega)$ — spectrum of original signal

&

$X_T(\omega)$ — spectrum of truncated signal

$$x_T[n] = x(n)\,w(n)$$

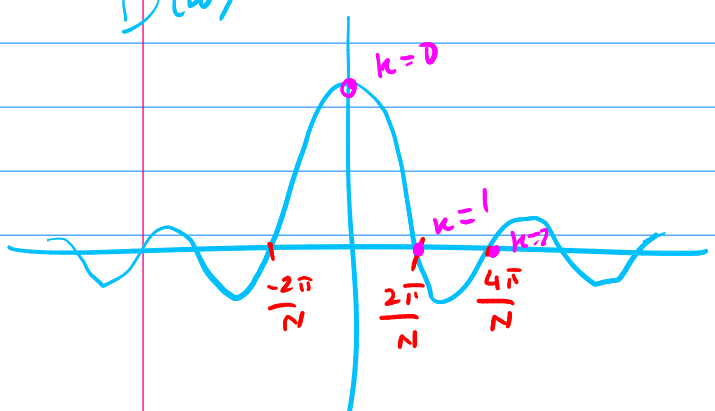
Window $w(n) = \begin{cases} 1 & ; \ 0 \le n \le N-1 \\ 0 & ; \ \text{else} \end{cases}$



Spectrum of $w(n)$:

$$W(\omega) = \frac{\sin\left(\frac{\omega N}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} e^{-j\omega\left(\frac{N-1}{2}\right)}$$

$D(\omega)$



$D(\omega)$

$D(\omega) \Rightarrow$ Dirichlet kernel

$$X_T(\omega) = \frac{1}{2\pi} \int_{\theta=-\pi}^{\pi} X(\theta)\, W(\omega-\theta)\, d\theta$$

Periodic convolution of $X(\omega)$ & $W(\omega)$



DFT Examples

① $x(n) = \begin{cases} 1 & ; \quad 0 \le n \le N-1 \\ 0 & ; \quad \text{else} \end{cases}$

$\{x(n)\} \xleftrightarrow[\text{DFT}]{N \text{ pt}} \{X[k]\}$ $\quad k = 0, \cdots, N-1$

$X[k] = \begin{cases} N & ; \quad k = 0 \\ 0 & ; \quad \text{else} \end{cases}$

② $x(n) = e^{j\frac{2\pi}{N}mn}$;

↓

one of bin freq. m^{th} bin freq.

$X[k] = \begin{cases} N & ; \quad k = m \\ 0 & ; \quad \text{else} \end{cases}$

③ $x(n) = e^{j\omega_\alpha n}$

$\omega_\alpha \neq$ bin freq.